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EDA CAPSTONE PROJECT

Data Science Job Salaries: What Drives Compensation?

SIC – SIC8 | Dataset: 565 records, 2020–2022

Compensation in data science is a black box for both sides of the hiring table.

FOR JOB SEEKERS

They don't know what to expect or negotiate for when evaluating a data science offer.

FOR COMPANIES

They risk over- or under-paying to compete for talent, without a clear picture of how compensation actually breaks down.

> The data science job market has grown rapidly, but there's still no clear picture of how pay breaks down by role, experience, company size, or location.

THE QUESTION

What actually determines a data scientist's salary — experience, company size, or location?

Seven questions shaped this analysis — and the dashboard built to answer them.

01 Does experience level significantly affect salary?

02 Does company size make a real difference in pay?

03 Do salaries increase over time (by work year)?

04 Does remote work raise salary?

05 Which job titles are the highest paying?

06 Does one country dominate the sample, and does that affect how we interpret the results?

07 How does employment type (Full-time, Contract, etc.) affect salary, and how much can that result be trusted?

A cleaned, well-structured panel — ready for analysis

Source: Kaggle "Data Science Job Salaries" dataset

SOURCE
Kaggle — Data Science Job Salaries

FINAL SHAPE
565 rows × 11 columns

TIME SPAN
2020 – 2022

DATA QUALITY
0 missing values · 0 duplicates

Originally 607 rows × 12 columns before cleaning.

KEY VARIABLES

experience_level	Entry / Mid / Senior / Executive
employment_type	FT / PT / CT / FL
job_title	(categorical)
salary_in_usd	(numeric, target)
employee_residence	(country code)
remote_ratio	0 / 50 / 100
company_location	(country code)
company_size	S / M / L

CHALLENGES → SOLUTIONS

CHALLENGE Raw file included a redundant leftover index column. → **SOLUTION** Dropped it during cleaning.

CHALLENGE 42 exact duplicate rows found in the raw export. → **SOLUTION** Removed via drop_duplicates, verified with a final 0-duplicate check.

CHALLENGE 10 extreme salary values (IQR outliers). → **SOLUTION** Investigated individually instead of deleting — most matched legitimate senior/executive roles.

From raw data to an interactive dashboard answering all seven questions.

A four-step pipeline — clean, explore, investigate, visualize.

01 **Clean the raw dataset**

Handle duplicates, verify data types, check for missing values.

02 **Run exploratory data analysis**

Distributions, group comparisons, correlation — to answer each business question.

03 **Investigate outliers instead of blindly removing them**

Verify each extreme value against its role and seniority context.

04 **Build an interactive Power BI dashboard**

So the findings can be filtered and explored live by any stakeholder.

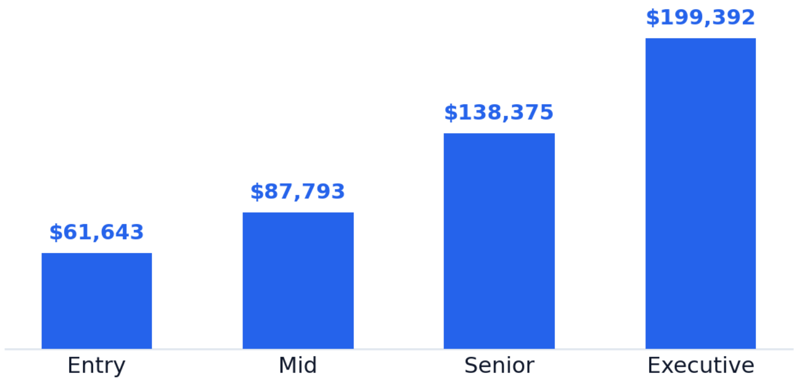
KEY TAKEAWAY

Experience level is the strongest driver of salary — **not company size**, **not location**.

Salaries scale sharply with experience — and 2022 outpaced prior years.

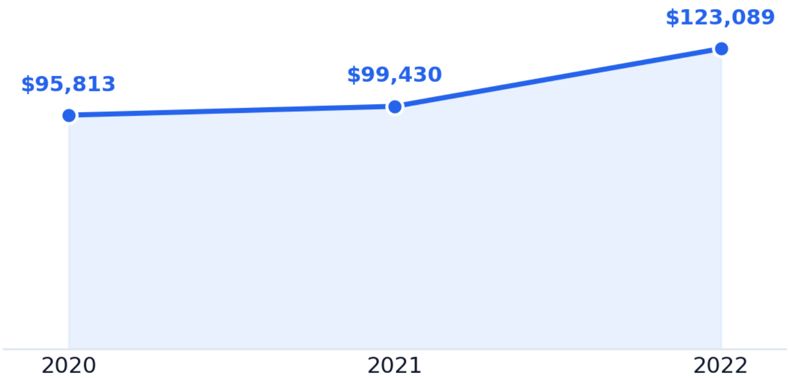
Avg. salary by experience level

USD, 2020–2022 mean



Avg. salary by year

USD, 2020–2022 mean

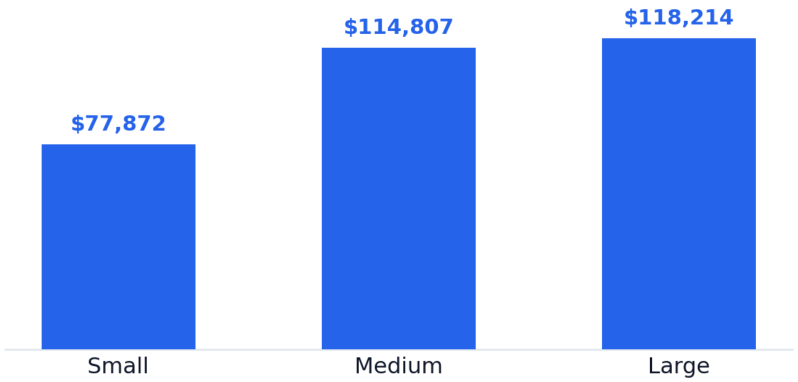


CALLOUT Salary roughly triples from Entry to Executive — a 3.2× jump.

Company size matters less than expected — and outliers are real.

Avg. salary by company size

USD, 2020–2022 mean



The gap between Medium and Large is smaller than expected.

OUTLIER ANALYSIS

10 potential outliers via IQR — **1.77%** of the dataset. Investigated, not deleted: most are legitimate senior or executive roles.

Right-skewed: bulk of salaries sit between $60K–$150K, with a long tail out to $600K.

HIGHEST SALARY OBSERVED

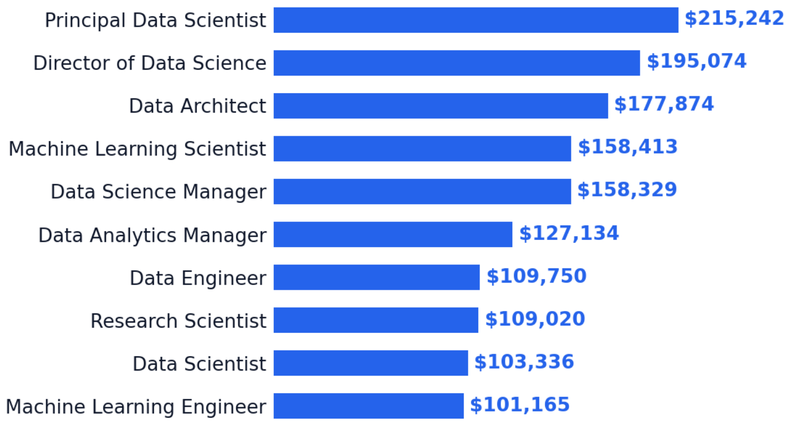
$600,000

Principal Data Engineer · Executive level · United States

The highest-paying roles and locations aren't evenly represented.

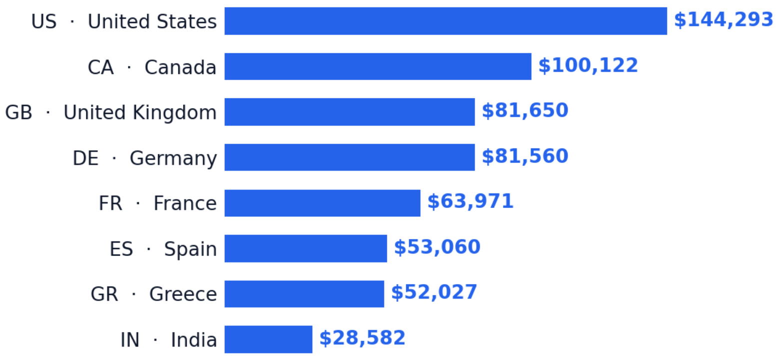
Top job titles by average salary

USD, 2020–2022 mean — roles with ≥ 7 observations



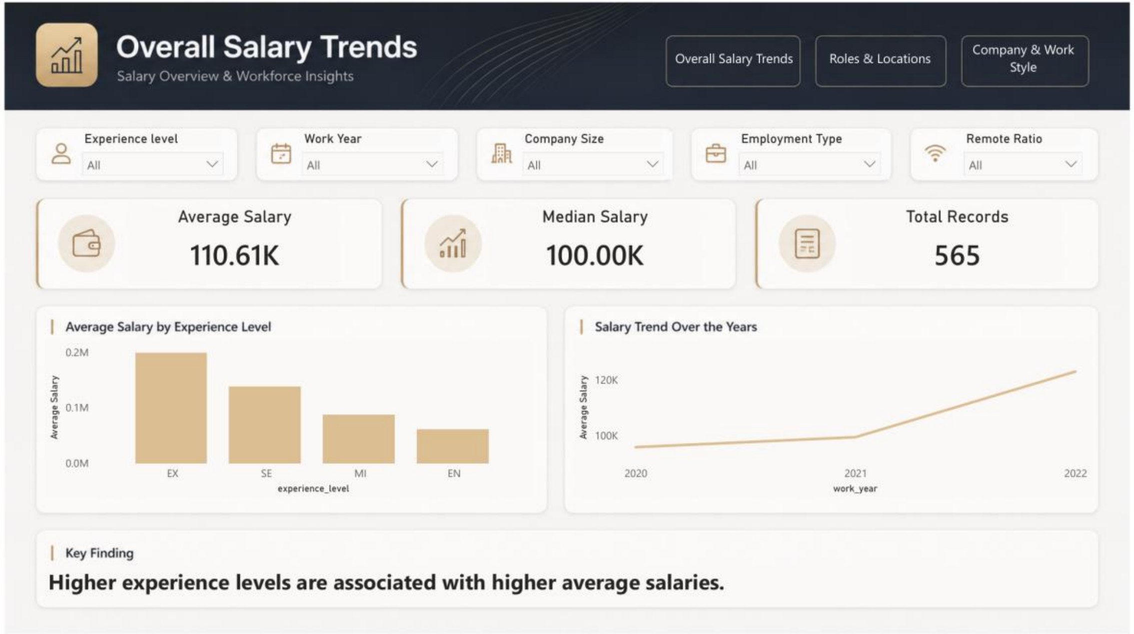
Top company locations by average salary

USD, 2020–2022 mean — US accounts for 318 of 565 records



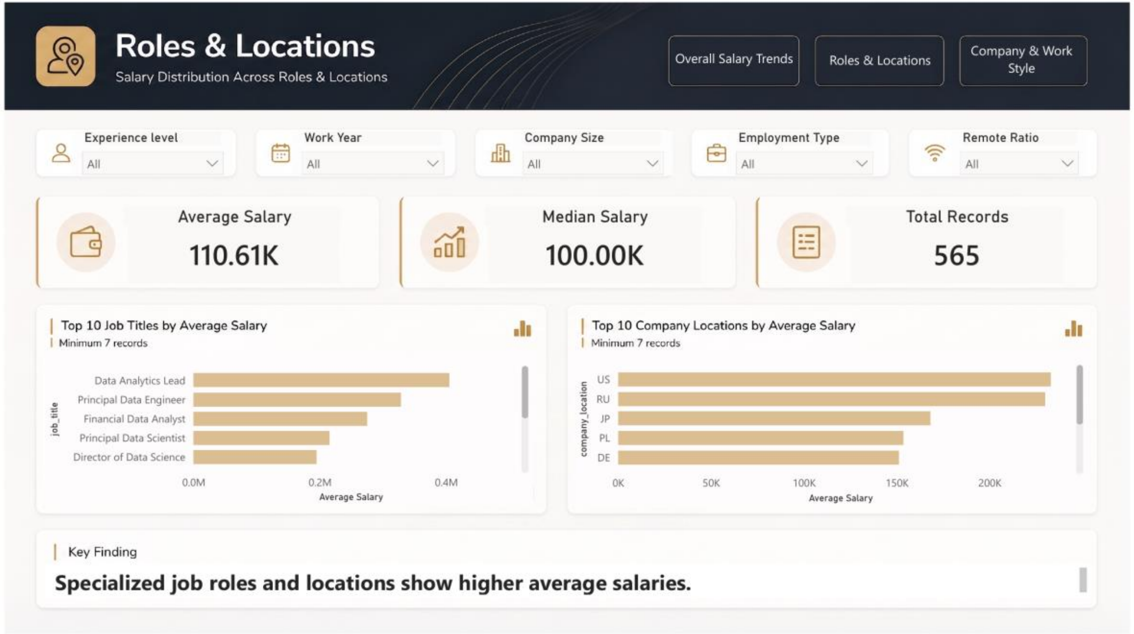
METHOD NOTE A minimum of 7 observations per group was used to filter out unstable, low-sample averages.

Overall Salary Trends



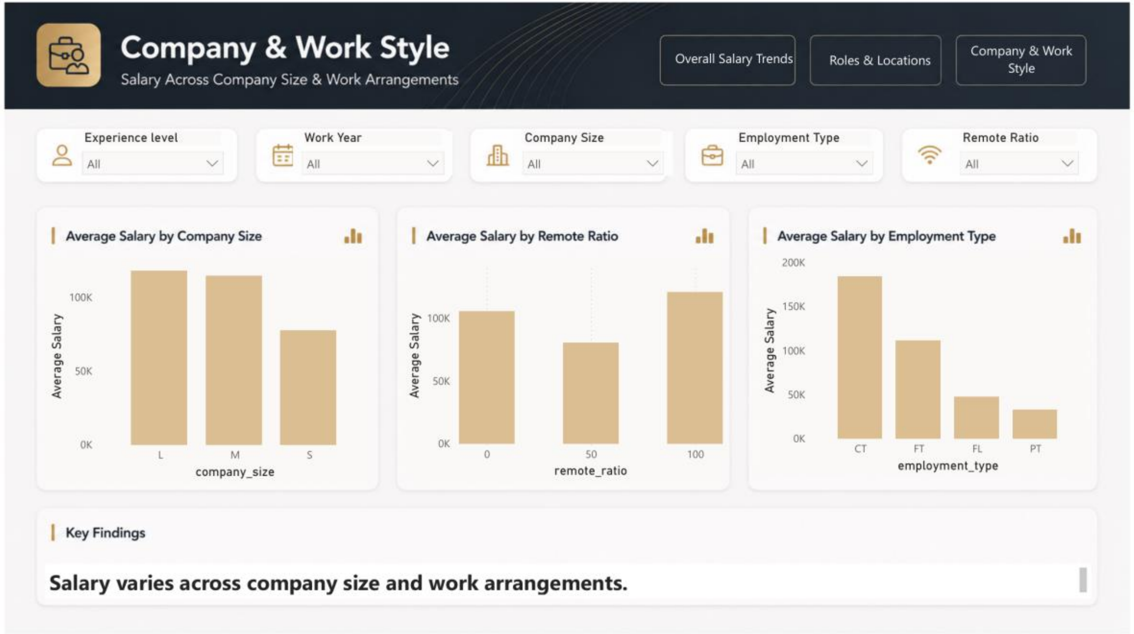
Interactive filters by experience level, work year, company size, employment type, and remote ratio.

Roles & Locations



Top 10 job titles and company locations by average salary, filtered live.

Company & Work Style



Salary breakdown by company size, remote ratio, and employment type.

Four actionable implications from the analysis.

Each card pairs the audience with a single directional guidance.

01

Job seekers

Benchmark against experience level, not company size.

02

Companies

Medium-sized firms can compete for senior talent without Large-company budgets.

03

Hiring managers

Treat employment-type differences (Contract > Full-time > Freelance > Part-
Part-time) as directional only — small sample sizes.

04

Remote work

Fully remote roles show the highest average salary (~$120,763) — but this is
is an association, not proof that remote work causes higher pay.

Read these findings with the appropriate caveats.

Four constraints bound how far the conclusions generalize.

- **US dominates the sample.**
  318 of 565 records (56%) come from the US — location comparisons should be read cautiously.

- **Small subgroups are statistically unstable.**
  Contract (5 records) and Freelance (4 records) are too thin to support strong claims.

- **Correlation tested only on numeric variables.**
  work_year vs. salary_in_usd: $r \approx 0.159$ — a weak positive relationship.

- **Only three years of data.**
  2020–2022 is too short a window to generalize the year-over-year trend.

Where the analysis goes from here.

Four follow-up directions to strengthen and extend the findings.

01

Extend the time window.

Add more recent years to test whether the 2022 salary jump holds or reverts.

02

Group by job category.

Add a job_category grouping for cleaner role-based comparisons across titles.

03

Rebalance geography.

Stratify the geographic analysis to reduce US dominance in the sample.

04

Validate remote-work effect.

Isolate the remote-work signal from confounders like role seniority and company size. size.

Answering the seven questions that shaped this project.

Q1 Experience level?	→	Yes — the strongest driver; salary roughly triples from Entry to Executive.
Q2 Company size?	→	Smaller effect than expected — Medium and Large pay similarly.
Q3 Salary over time?	→	Yes — average salary rose from ~$95.8K (2020) to ~$123.1K (2022).
Q4 Remote work?	→	Highest average pay (~$120.8K), but an association, not proof of causation.
Q5 Highest-paying titles?	→	Principal Data Scientist (~$215K), Director of Data Science (~$195K).
Q6 One country dominating?	→	Yes — the US is 56% of the sample; location comparisons need caution.
Q7 Employment type?	→	Contract shows the highest average, but the sample is too small (5 records) to trust firmly.

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Thank You.

Questions?

Experience level remains the strongest driver of compensation in data science roles.